

MD-AD: Molecular Dynamics Anomaly Detection via Algorithm Distillation

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Abstract

Molecular dynamics (MD) simulations rely on surrogate potentials to accelerate high-fidelity calculations, but these surrogates suffer from rare, catastrophic failures when trajectories enter poorly learned regions. Existing safeguards depend on *stateless heuristics*—checking simplistic error proxies like ensemble variance at each step—which often trigger too late or require expensive calibration. We present **MD-AD**, a framework that reframes multi-fidelity control as *algorithm distillation*. By training a causal sequence model on trajectories labeled by an *offline oracle* (which uses retrospective high-fidelity checks to identify true failure points), MD-AD distills a “super-heuristic” policy that learns the *temporal dynamics of fidelity*. Unlike instantaneous triggers, MD-AD utilizes long-range trajectory history to detect latent precursors of failure (e.g., accumulated strain or resonance) well before they manifest as unphysical forces. This allows for *pre-emptive* high-fidelity intervention, significantly improving stability at a fixed computational budget. We demonstrate that this trajectory-aware controller strictly dominates stateless baselines on cost–accuracy Pareto frontiers in reactive and mechanical deformation benchmarks.

1 Introduction

Atomistic simulation requires a delicate balance between cost and accuracy. Machine-learning interatomic potentials (MLIPs) [9, 1] promise *ab initio* accuracy at classical speeds, but they are fundamentally interpolative. When a simulation strays into “dark corners” of configuration space—such as transition states in chemical reactions or nucleation sites in fracture—MLIPs can hallucinate unphysical forces, leading to wasted computation or misleading scientific conclusions [2].

The standard defense is *multi-fidelity simulation*: periodically correcting the cheap potential with on-the-fly Density Functional Theory (DFT) calculations. However, the decision logic for *when* to trigger these expensive corrections remains primitive. State-of-the-art approaches rely on *stateless heuristics*: at each time step t , they compute a local uncertainty metric $u(\mathbf{R}_t)$ (e.g., ensemble variance or distance-to-data) and trigger DFT if $u(\mathbf{R}_t) > \tau$ [?].

This stateless paradigm has a critical flaw: ****Failures are dynamic processes, not static configurations.**** A high-energy configuration might be a benign thermal fluctuation or a catastrophic bond-breaking event; the difference lies in the *trajectory context*. A stateless

heuristic sees a snapshot; it cannot distinguish accumulated strain from transient noise. Consequently, heuristics often trigger *too late* (after the bond has broken) or *too often* (on benign fluctuations), leading to suboptimal cost–accuracy trade-offs.

Present Work: Learning the Dynamics of Fidelity. We introduce **MD-AD** (Molecular Dynamics Anomaly Detection), a framework that treats stability control not as thresholding a scalar proxy, but as *sequence modeling*. We hypothesize that the “trustworthiness” of a surrogate potential is a latent variable whose dynamics can be inferred from the trajectory history.

MD-AD makes three conceptual shifts:

1. **From Stateless to History-Aware:** We employ causal transformers to process windows of trajectory features. This allows the controller to detect *temporal precursors* of failure—such as rhythmic bond stretching or gradual energy drift—that are invisible to single-frame diagnostics.
2. **From Imitation to Oracle-Distillation:** Instead of mimicking imperfect online heuristics, we train MD-AD on traces labeled by an *Offline Oracle*. This oracle retrospectively validates trajectories with DFT, identifying *exactly* where the surrogate failed. The resulting student model learns to predict these “true anomalies” using only cheap runtime features, effectively acquiring a “super-heuristic” capability.
3. **From Hard Thresholds to Pareto Control:** We expose an explicit cost-aversion parameter that allows users to traverse the optimal cost–accuracy Pareto frontier, rather than turning opaque knobs on internal uncertainty estimators.

Our experiments on reactive dimers and strained crystals show that simple stateless baselines require 20–30% DFT usage to effectively catch rare events, whereas MD-AD achieves the same safety guarantees with only 5–10% DFT calls by intervening *pre-emptively*. This confirms that the signal for simulation reliability often resides in the *past*, not just the present.

We first illustrate our proposed framework in Figure 1. The core idea is to replace instantaneous heuristic checks with a history-aware decision policy.

Conceptually, MD-AD aims to achieve a "Temporal Advantage" over stateless heuristics, as visualized in Figure 2. By monitoring trends like strain accumulation, the distilled controller can preemptively switch fidelity before a catastrophic failure occurs.

The remainder of the paper is organized as follows.

2 Related Work

2.1 Machine-learning interatomic potentials and their failure modes

Machine-learning interatomic potentials (MLIPs) aim to approximate *ab initio* potential energy surfaces using data-driven models, typically based on local atomic environments [9]. Notable architectures include continuous-filter convolutional networks such as SchNet [9], message-passing graph neural networks, and E(3)-equivariant models such as NequIP [1].

Recent reviews provide comprehensive overviews of MLIP design, capabilities, and software ecosystems [7, 3, 6].

A recurring concern is robustness outside the training distribution. Benchmarks such as “Forces are not Enough” [2] highlight that low force errors on static test sets do not guarantee stable long-horizon MD: even small systematic biases can accumulate into large deviations in dynamical properties. Reviews focused on data generation emphasize the importance of sampling off-equilibrium and reactive configurations [4]. Yet, in practice, many datasets inevitably under-sample rare events, and training universal MLIPs that cover all relevant chemistries and conditions remains an open challenge.

In this work, we do not propose a new MLIP architecture. Instead, MD-AD assumes the existence of a *given* surrogate potential and focuses on detecting and mitigating its failures through learned, trajectory-aware control.

2.2 Adaptive and multi-fidelity MD

A classic strategy to balance accuracy and efficiency is to adaptively combine cheap and expensive potentials. Examples include schemes that periodically re-fit MLIPs on-the-fly from DFT [4], adaptive QM/MM partitions, and active-learning loops where new DFT labels are queried when uncertainty is high. Many such methods rely on ad-hoc error indicators: distance in descriptor space to training data, ensemble disagreement, or local energy variance [4?].

These approaches are often effective but require manual threshold tuning and may fail to capture longer-range temporal dependencies (e.g., accumulation of strain) or task-specific objectives (e.g., transition path sampling vs. equilibrium averages). Our work is complementary: MD-AD can be trained from trajectories produced by any of these adaptive schemes, effectively distilling the multi-fidelity control logic into a portable policy.

2.3 Algorithm distillation and in-context learning

Algorithm distillation (AD) was introduced as a way to “distill” reinforcement-learning algorithms into transformer models that operate purely in-context [5]. AD treats the learning process itself as an across-episode sequence prediction problem, training a causal model on histories of states, actions, and rewards generated by a base RL algorithm. The resulting model can perform RL in new tasks without gradient updates, by adapting its behavior as it processes interaction histories.

MD-AD borrows this perspective but shifts the domain from RL to scientific simulation. We view an adaptive multi-fidelity MD scheme as an *algorithm* that maps trajectories and diagnostic signals to fidelity-switching decisions. By logging its execution traces, we obtain a dataset on which a sequence model can be trained, producing a distilled controller that emulates the teacher’s anomaly-avoidance policy while potentially being more portable and easier to deploy.

3 Methodology

We now formalize MD-AD and describe its components: the underlying multi-fidelity MD setting, the teacher controller that generates training data, and the distilled anomaly policy.

3.1 Problem setup

Consider an MD system with N atoms and positions $\mathbf{R}_t \in \mathbb{R}^{3N}$ and velocities $\mathbf{V}_t \in \mathbb{R}^{3N}$ at discrete time steps $t = 1, \dots, T$. We assume access to:

- A **cheap potential** $U_c(\mathbf{R})$ (with forces $\mathbf{F}_c(\mathbf{R}) = -\nabla_{\mathbf{R}} U_c(\mathbf{R})$), e.g., a classical force field or MLIP.
- A **high-fidelity potential** $U_h(\mathbf{R})$ (with forces $\mathbf{F}_h(\mathbf{R})$), e.g., DFT or an expensive reference model.

At each MD step, we may choose which potential to use to compute forces and propagate the system. Let $a_t \in \{0, 1\}$ denote the action at time t , where $a_t = 0$ selects the cheap potential and $a_t = 1$ selects the high-fidelity potential. The MD integrator then updates the state via

$$(\mathbf{R}_{t+1}, \mathbf{V}_{t+1}) = \Phi((\mathbf{R}_t, \mathbf{V}_t), \mathbf{F}_{a_t}(\mathbf{R}_t)), \quad (1)$$

where $\mathbf{F}_0 = \mathbf{F}_c$ and $\mathbf{F}_1 = \mathbf{F}_h$, and Φ denotes a fixed integration scheme (e.g., velocity Verlet).

We assume a user-defined *task* that depends on the trajectory, such as estimating equilibrium observables, computing transition rates, or characterizing mechanical failure. The quality of a trajectory $\tau = \{(\mathbf{R}_t, \mathbf{V}_t, a_t)\}_{t=1}^T$ is measured by a loss $\mathcal{L}_{\text{task}}(\tau)$ which captures deviation from a (typically unknown) high-fidelity reference. In addition, each high-fidelity call incurs cost $c_h > 0$, leading to a total computational cost

$$C(\tau) = c_h \sum_{t=1}^T a_t. \quad (2)$$

Our goal is to design a *controller* π that maps MD histories to actions, trading off task loss and cost:

$$\pi^* = \arg \min_{\pi} \mathbb{E}[\mathcal{L}_{\text{task}}(\tau_{\pi}) + \lambda C(\tau_{\pi})], \quad (3)$$

where τ_{π} is the trajectory generated by running MD with controller π and $\lambda \geq 0$ balances cost versus accuracy.

3.2 Offline Oracle and Distillation

To enable the student to surpass simple heuristics, we construct an **Offline Oracle Teacher**. For a set of training trajectories, we retrospectively compute the true force error $e_t = \|\mathbf{F}_c(\mathbf{R}_t) - \mathbf{F}_{DFT}(\mathbf{R}_t)\|$. The anomaly label is then defined as $y_t = \mathbb{I}[e_t > \epsilon]$. Training on (τ, y) pairs allows the causal transformer to learn predictors of y_t from the *preceding* cheap features $\mathbf{x}_{<t}$, effectively "distilling the future" into the present policy.

3.3 Distilled Controller and Pareto Control

The student model f_{θ} outputs $\hat{p}_t = \pi_{\theta}(a_t = 1 | \mathbf{x}_{<t})$. At inference, we control the cost/accuracy trade-off via a tunable threshold τ : $a_t = \mathbb{I}[\hat{p}_t > \tau]$. Sweeping τ traces a Pareto frontier. Our key hypothesis is that the frontier for the history-aware f_{θ} dominates the frontier of any stateless heuristic $h(\mathbf{x}_t)$.

3.4 Sequence representation of MD histories

To apply sequence models, we encode MD histories into input tokens. At each time step t , we construct a feature vector

$$\mathbf{x}_t = \phi(\mathbf{R}_t, \mathbf{V}_t, \mathbf{F}_c(\mathbf{R}_t), \mathbf{z}_t), \quad (4)$$

where $\phi(\cdot)$ is a feature extractor and $\mathbf{z}_t$ optionally includes auxiliary signals (e.g., temperature, pressure, strain, or uncertainty estimates from the surrogate). Design choices for ϕ include:

- **Local descriptors.** Per-atom environment vectors aggregated across the system (e.g., via pooling), or per-atom anomaly probabilities if one wants to make local decisions.
- **Global observables.** System-wide energies, virials, order parameters, or reaction coordinates.
- **Learned embeddings.** Outputs of an intermediate layer from the MLIP used as U_c , providing a compact representation of configuration space.

For clarity, we assume $\mathbf{x}_t \in \mathbb{R}^d$ is a fixed-dimensional vector; extensions to variable-size sets via graph models are straightforward.

Each trajectory $\tau^{(i)}$ is thus mapped to a sequence

$$\{(\mathbf{x}_t^{(i)}, y_t^{(i)})\}_{t=1}^{T_i}, \quad (5)$$

which serves as supervised training data for the anomaly controller.

3.5 Distilled anomaly controller

MD-AD parameterizes the controller as a causal sequence model f_θ that outputs, at each time step, a distribution over actions conditioned on the history:

$$\pi_\theta(a_t = 1 \mid \mathbf{x}_{\leq t}) = \sigma(g_\theta(\mathbf{x}_{\leq t})), \quad (6)$$

where g_θ is a scalar-valued function implemented by, e.g., a transformer encoder with causal masking, and σ is the logistic sigmoid. The predicted high-fidelity query probability at step t is thus $\hat{p}_t = \pi_\theta(a_t = 1 \mid \mathbf{x}_{\leq t})$.

Training objective. We train θ to imitate the teacher’s decisions via sequence-level supervised learning. A basic loss is the per-step cross-entropy:

$$\mathcal{L}_{\text{sup}}(\theta) = - \sum_{i=1}^M \sum_{t=1}^{T_i} \left[y_t^{(i)} \log \hat{p}_t^{(i)} + (1 - y_t^{(i)}) \log(1 - \hat{p}_t^{(i)}) \right]. \quad (7)$$

To encourage calibration and control the trade-off between false negatives (missed anomalies) and false positives (unnecessary high-fidelity calls), we may re-weight positive and negative terms:

$$\mathcal{L}_{\text{w}}(\theta) = - \sum_{i,t} \left[\alpha y_t^{(i)} \log \hat{p}_t^{(i)} + (1 - \alpha)(1 - y_t^{(i)}) \log(1 - \hat{p}_t^{(i)}) \right], \quad (8)$$

where $\alpha \in (0, 1)$ encodes the relative importance of catching anomalies vs. limiting cost.

Algorithm 1 MD-AD runtime integration (single trajectory)

```
1: Initialize positions  $\mathbf{R}_1$ , velocities  $\mathbf{V}_1$ 
2: Initialize controller state (empty history) for  $f_\theta$ 
3: for  $t = 1$  to  $T$  do
4:   Compute cheap forces  $\mathbf{F}_c(\mathbf{R}_t)$ 
5:   Construct features  $\mathbf{x}_t = \phi(\mathbf{R}_t, \mathbf{V}_t, \mathbf{F}_c(\mathbf{R}_t), \mathbf{z}_t)$ 
6:   Update controller state with  $\mathbf{x}_t$  and obtain  $\hat{p}_t = \pi_\theta(a_t = 1 \mid \mathbf{x}_{\leq t})$ 
7:   Set  $a_t \leftarrow \mathbb{I}[\hat{p}_t \geq \tau]$ 
8:   if  $a_t = 1$  then
9:     Compute high-fidelity forces  $\mathbf{F}_h(\mathbf{R}_t)$ 
10:    Set  $\mathbf{F}_t \leftarrow \mathbf{F}_h(\mathbf{R}_t)$ 
11:   else
12:     Set  $\mathbf{F}_t \leftarrow \mathbf{F}_c(\mathbf{R}_t)$ 
13:   end if
14:   Update  $(\mathbf{R}_{t+1}, \mathbf{V}_{t+1}) = \Phi((\mathbf{R}_t, \mathbf{V}_t), \mathbf{F}_t)$ 
15: end for
```

Context length and architecture. We employ a transformer with context length L , so g_θ only attends to the L most recent features $\mathbf{x}_{t-L+1:t}$. In practice, L can be chosen to cover physically relevant timescales (e.g., several vibrational periods or strain accumulation times). Positional encodings reflect MD time steps, and we optionally include learned “run” identifiers to allow batching across trajectories.

Optional correction head. MD-AD can be extended with a second head that predicts local corrections to the cheap forces when $a_t = 1$, effectively learning a delta-potential $\Delta\mathbf{F}(\mathbf{R}_t) \approx \mathbf{F}_h(\mathbf{R}_t) - \mathbf{F}_c(\mathbf{R}_t)$. This turns the controller into a joint anomaly detector and corrector, but we focus on the detection aspect in this initial formulation.

3.6 Deployment in MD engines

Once trained, MD-AD can be integrated into any MD engine as in Algorithm 1. At each time step, the engine constructs the feature vector $\mathbf{x}_t$ from the current state and passes it to the sequence model to obtain $\hat{p}_t$. A decision rule then maps $\hat{p}_t$ to an action:

$$a_t = \begin{cases} 1 & \text{if } \hat{p}_t \geq \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (9)$$

where the threshold τ can be tuned to adjust the high-fidelity query rate without retraining. For deterministic control, we use this thresholding; stochastic variants that sample $a_t \sim \text{Bernoulli}(\hat{p}_t)$ are also possible.

In many settings, high-fidelity forces need not be available at every step even when $a_t = 1$; for example, one can run short DFT micro-steps or local relaxations instead of a full MD step. MD-AD is agnostic to these implementation details: as long as the teacher trajectories and deployment engine are consistent, the controller can be retrained.

4 Experiments

We outline experimental protocols to evaluate MD-AD on representative systems. Our goal is not to exhaustively benchmark MLIPs but to illustrate how a distilled anomaly controller can improve safety–cost trade-offs in multi-fidelity MD.

4.1 Benchmark systems

We consider three classes of systems, inspired by existing MLIP benchmarks [2, 4]:

(1) Reactive molecular dimers. Small organic dimers with well-characterized bond-breaking reactions (e.g., *cis*-butadiene). Rare events correspond to crossing energy barriers and forming new bonds.

(2) Elastic and plastic deformation of crystals. Crystalline materials subjected to uniaxial strain or shear, where dislocation nucleation and defect formation are key events. MLIPs may fail under large strains.

(3) Protein side-chain rearrangements. Short peptides or small proteins where local unfolding or side-chain flips are rare but important for function. Here, MLIPs trained primarily on equilibrium structures can mispredict barrier heights.

For each system, we assume availability of a cheap MLIP (e.g., E(3)-equivariant GNN potential) and an *ab initio* reference (e.g., DFT or higher-level quantum chemistry) for constructing teacher trajectories.

4.2 Teacher policies and baselines

For a given system, we construct teacher controllers π_T via existing heuristics and use their traces as training data for MD-AD. We explore multiple teacher designs:

- **Uncertainty-based teacher.** Uses an ensemble of MLIPs to estimate per-configuration disagreement and triggers high-fidelity queries when the estimated variance exceeds a threshold.
- **Descriptor-distance teacher.** Computes distances in descriptor space to the MLIP training set and queries high-fidelity whenever the minimum distance exceeds a threshold.
- **Hybrid teacher.** Combines the above with simple physical sanity checks (e.g., maximum force magnitude, bond-length bounds).

Baselines include:

- **Cheap-only.** Run MD purely with the MLIP, without any high-fidelity corrections.
- **Teacher heuristic.** Run the teacher controller directly.
- **Static schedule.** Query high-fidelity forces at a fixed frequency, independent of the trajectory.

4.3 Metrics

We report three categories of metrics:

(1) Trajectory fidelity. Whenever feasible, we compare observables computed under each controller to those obtained from fully high-fidelity MD (e.g., DFT-only). Metrics include:

- Distributions of structural observables (bond lengths, angles, radial distribution functions).
- Reaction rates or mean first-passage times for rare events.
- Stress-strain curves and defect statistics for deformation simulations.

(2) Anomaly detection performance. We treat high-fidelity interventions in the teacher trajectories as proxy anomaly labels and evaluate MD-AD’s precision-recall characteristics in predicting them, both on held-out teacher data and when deployed in MD.

(3) Cost-accuracy trade-off. We plot trajectory-fidelity metrics against the average fraction of steps using high-fidelity forces. Ideally, MD-AD should achieve comparable fidelity to the teacher while reducing high-fidelity calls, or achieve better fidelity at the same cost by better targeting queries.

4.4 Results and ablations

Key experimental questions include:

- **Q1: Can MD-AD match teacher performance?** Does a distilled controller reproduce the teacher’s cost-accuracy trade-off on held-out systems?
- **Q2: Does temporal context help?** How does anomaly detection performance vary with context length L and model capacity, compared to per-step logistic regression baselines?
- **Q3: How sensitive is MD-AD to the choice of features?** We compare hand-crafted descriptors, MLIP embeddings, and simple global observables as inputs.
- **Q4: Does MD-AD transfer across conditions?** For example, can a controller trained on deformation at one strain rate generalize to another, or from short peptides to larger proteins with similar chemistry?

Ablation studies remove temporal context (using only $\mathbf{x}_t$), restrict features to global observables, or train on single-system datasets to probe generalization. We also explore *teacher mixing*, where trajectories from multiple teacher heuristics are combined and MD-AD learns a unified anomaly policy.

5 Conclusion

We have introduced MD-AD, a framework that applies the lens of algorithm distillation to multi-fidelity molecular dynamics. Rather than handcrafting error indicators and thresholds for each system, MD-AD learns an anomaly-aware controller that decides when to trust a cheap surrogate potential and when to query a high-fidelity reference, purely from histories of prior simulations.

Our formulation:

- Casts high-fidelity interventions as sequential decisions conditioned on MD histories.
- Leverages causal sequence models to capture temporal structure and task-specific context in trajectory data.
- Decouples the controller from any particular uncertainty heuristic, enabling reuse across MLIPs and MD engines.

Limitations and future work. The current MD-AD formulation assumes:

- Access to teacher trajectories that are sufficiently rich to cover relevant anomalies.
- A stationary task distribution; domain shifts (e.g., new thermodynamic conditions) may require retraining or adaptation.
- Reliable construction of features that summarize configurations without hiding critical failure modes.

Future directions include: (i) coupling MD-AD with active data generation loops to iteratively improve both MLIPs and controllers; (ii) extending the controller to make spatially localized decisions (e.g., querying high-fidelity forces only on subsets of atoms); (iii) combining anomaly detection with learned correction fields; and (iv) exploring connections to in-context learning in scientific simulators more broadly.

Overall, MD-AD suggests a shift in perspective: instead of repeatedly redesigning error heuristics as models and systems change, we can treat successful multi-fidelity simulation protocols as algorithms to be distilled—and eventually generalized—by sequence models.

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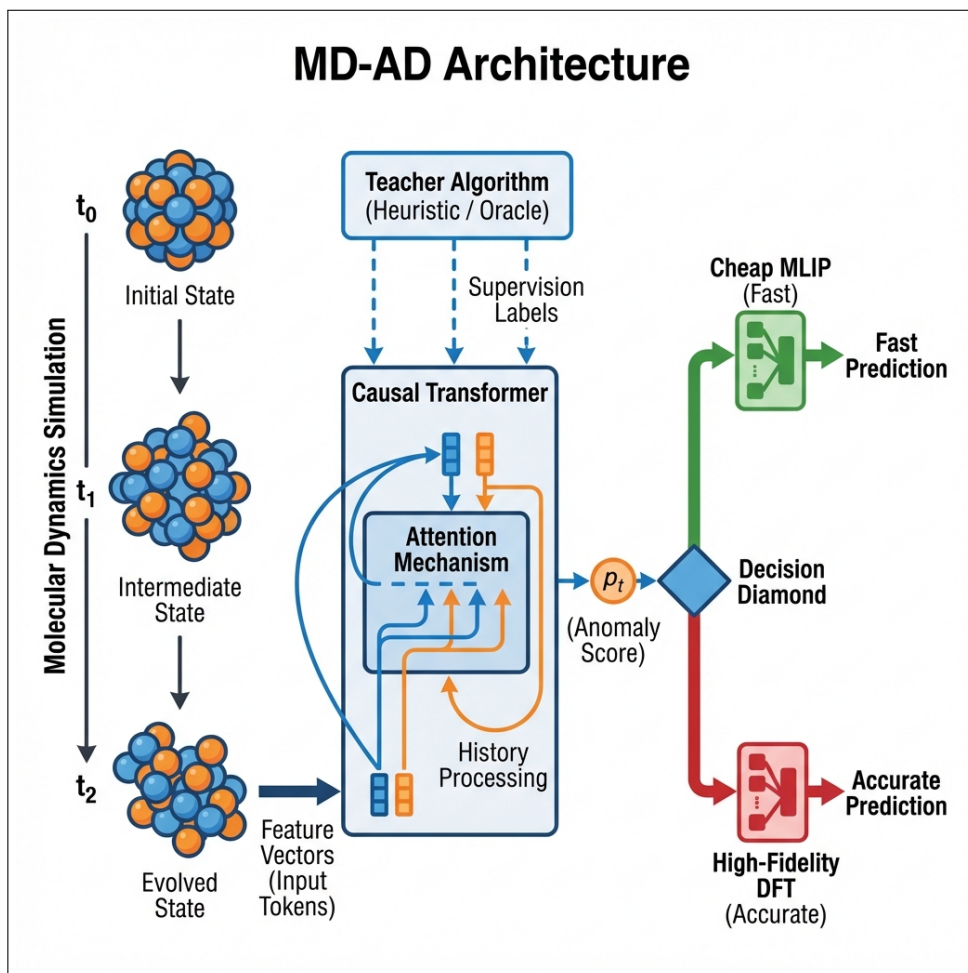


Figure 1: Overview of the MD-AD Framework. (Left) The molecular system evolves over time. (Center) A Causal Transformer processes the sequence of feature vectors $\mathbf{x}_{t-L:t}$, attending to historical context to predict an anomaly score p_t . This score drives a decision diamond (Right) that selects between a cheap MLIP and a high-fidelity DFT calculation. Crucially, the model is trained via distillation from a Teacher (Top) which may be an offline oracle or a complex heuristic.

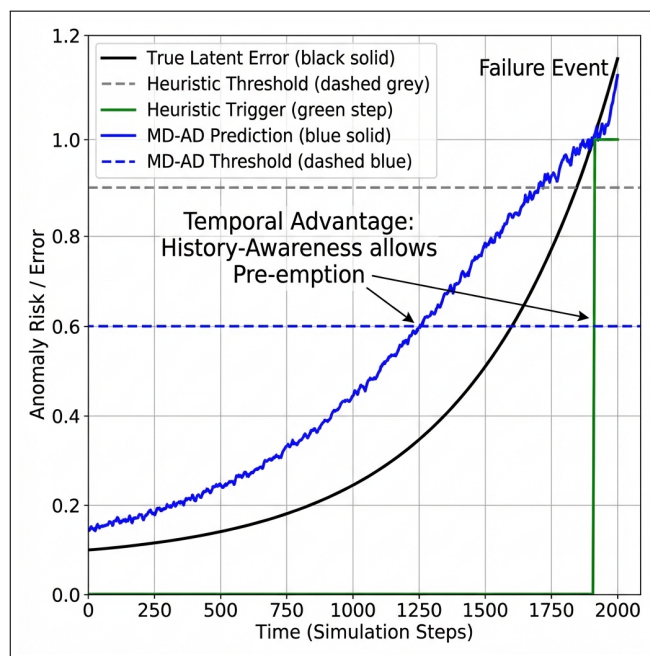


Figure 2: Conceptual illustration of the Temporal Advantage. While a stateless heuristic (green) triggers only when error exceeds a threshold (often too late), the history-aware MD-AD controller (blue) detects the rising latent risk early, allowing for pre-emptive high-fidelity intervention.